Calculus Honors - Teacher Manual

Semester 2: Applications of the Derivative, then Integrals - Florida Course 1202300

PURPOSE OF THIS MANUAL

This manual equips a teacher to run Semester 2 of Calculus Honors (Quarters 3 and 4, Modules 9 through 16, and finals week) as an asynchronous Canvas course that a live class can also use unchanged. It covers the weekly rhythm, the assessment architecture and your time budget, module-by-module teaching notes with standards alignment, where the answer keys are, and the finals week. Standards codes appear here for your reference; they never appear on student-facing pages. The Semester 1 manual (Quarter 1) and the Rubrics document apply to the whole year.

THE CALENDAR

Module	Weeks	Title
9	18-19	Tangent Lines and Linear Approximation
10	20-21	Increasing, Decreasing, and Extrema
11	22-23	Concavity and Curve Sketching
12	24-26	Optimization, Motion, and Related Rates
13	27-28	Riemann Sums and Approximating Integrals
14	29-30	The Fundamental Theorem of Calculus
15	31-32	Antiderivatives and Differential Equations
16	33-34	Area, Average Value, and Volume
Finals	35	Semester 2 review lesson, review activity, and final exam

Course weeks continue from Semester 1 (Weeks 1 to 17), so Semester 2 runs course Weeks 18 to 35. Module 12 runs three weeks; every other module runs two. In the date panel of the Course Kit, the semester starts at its Week 1, which corresponds to course Week 18.

THE WEEKLY RHYTHM

- **Lessons.** Usually two per week (one each in Weeks 25 and 26). Interactive pages with worked examples, self-check widgets, and diagrams. Ungraded; students work the practice in a notebook.
- **One problem set per week** (10 points). Seven problems worked by hand and uploaded as one PDF or image file; scored on completion and good-faith effort.
- **One auto-graded quiz per week** (8 questions: four multiple choice, four short answer). Answer keys are in each quiz. Short-answer questions accept equivalent forms (for example 1/2 and 0.5).
- **Per module:** a module map, an opening reflection (survey), a discussion (ungraded, one post and one reply), a Module Challenge (16 points, four-criterion rubric), and a closing reflection (survey).
- **No VR.** Semester 2 has no VR activities, by decision while the VR app is being rebuilt.

ASSESSMENT ARCHITECTURE AND YOUR TIME BUDGET

Tier	Items	Teacher time
1 - Material (auto)	Weekly quizzes, weekly problem sets (completion), in-lesson widgets	~0 hrs
2 - Community	Module discussion, reflection triage	~2 to 3 hrs/week

Tier	Items	Teacher time
3 - Teacher	Module Challenge (four-criterion rubric); Semester Review Activity and Semester Final in finals week	~8 to 12 hrs/module cycle; the final is the semester's one large grading load
4 - Scaffold	Module map, opening and closing reflections	~15 to 20 min/module

Mine the reflection surveys: the confidence item is your early-warning signal, and the "what remains unclear" field tells you where to drop public comments in the discussion.

MODULE-BY-MODULE TEACHING NOTES

Module 9: Tangent Lines and Linear Approximation (Weeks 18 to 19)

Standards: MA.912.C.3.1, MA.912.C.3.2. **Essential question:** What can a tangent line tell you about a curve near one point, and how far can you trust it?

Lessons: 9.01 equations of tangent and normal lines; 9.02 vertical tangents and cusps; 9.03 linear approximation; 9.04 differentials and error estimation.

Watch for: Students treat Δy and dy as the same thing, and read a differential estimate as a bound rather than an estimate. The vertical tangent (Lesson 9.02) is an edge case the standards name explicitly; make sure students can tell it from a cusp and a corner.

Module 10: Increasing, Decreasing, and Extrema (Weeks 20 to 21)

Standards: MA.912.C.3.3, MA.912.C.3.4. **Essential question:** How does the derivative tell you where a function climbs, falls, and turns?

Lessons: 10.01 critical points and local extrema; 10.02 increasing, decreasing, and the First Derivative Test; 10.03 absolute extrema on a closed interval; 10.04 reading extrema from graphs of f and f-prime.

Watch for: A critical point is a candidate, not a verdict (x cubed has one and no extremum). On closed intervals insist students compare endpoint values. A peak of the graph of f-prime is not an extremum of f.

Module 11: Concavity and Curve Sketching (Weeks 22 to 23)

Standards: MA.912.C.3.5, MA.912.C.3.6. **Essential question:** What does the second derivative add, and how do f, f' and f'' fit into one picture?

Lessons: 11.01 concavity and the second derivative; 11.02 inflection points and the Second Derivative Test; 11.03 sketching a curve from its derivatives; 11.04 matching f, f-prime, and f-double-prime.

Watch for: f''(c) = 0 does not by itself give an inflection point (x to the fourth); require the sign change. The Second Derivative Test is silent when f''(c) = 0, so students need the First Derivative Test as a fallback.

Module 12: Optimization, Motion, and Related Rates (Weeks 24 to 26)

Standards: MA.912.C.3.7, MA.912.C.3.9, MA.912.C.3.10. **Essential question:** How do you turn a story about the world into a function, and then let calculus find the best case or the rate of change?

Lessons: Week 24: 12.01 applied optimization, setting up the problem; 12.02 applied optimization, geometry, cost, and distance. Week 25: 12.03 motion along a line. Week 26: 12.04 related rates.

Watch for: This is the heaviest applications module and runs three weeks (one lesson in each of Weeks 25 and 26). In optimization, check that students justify the optimum (closed-interval method or sign of f-prime), state the domain, and answer in context with units. In related rates, the common error is substituting a number before differentiating with respect to time.

Module 13: Riemann Sums and Approximating Integrals (Weeks 27 to 28)

Standards: MA.912.C.4.1, MA.912.C.4.2. **Essential question:** How can you add up infinitely many infinitely thin pieces and get an exact answer?

Lessons: 13.01 approximating area with rectangles; 13.02 sigma notation and Riemann sums; 13.03 the definite integral; 13.04 the Trapezoidal Rule.

Watch for: Left versus right endpoints (which over- or underestimates depends on whether the function is increasing or decreasing). The lessons write sums with index i and points x sub i; the Week 27 problem set uses k, which is only a placeholder. The error bound in the Trapezoidal Rule lesson is stated without proof, and the lesson says so.

Module 14: The Fundamental Theorem of Calculus (Weeks 29 to 30)

Standards: MA.912.C.4.3 - MA.912.C.4.7. **Essential question:** Why does finding an area come down to undoing a derivative?

Lessons: 14.01 the Fundamental Theorem, Part 1; 14.02 the Fundamental Theorem, Part 2; 14.03 net change and analyzing graphs with the theorem; 14.04 integration by substitution.

Watch for: Dense module; watch pacing. For Part 1 with a variable upper limit, require the chain-rule factor. In definite substitution, students either change the limits or return to x; mixing the two is the standard error. Lesson 14.04 shows when substitution does not apply.

Module 15: Antiderivatives and Differential Equations (Weeks 31 to 32)

Standards: MA.912.C.5.1 - MA.912.C.5.4. **Essential question:** How do you recover a function from information about how it changes?

Lessons: 15.01 antiderivatives and initial conditions; 15.02 differential equations and slope fields; 15.03 separable differential equations; 15.04 exponential growth and decay.

Watch for: Do not let students drop the constant of integration before applying an initial condition. In separation of variables, a constant solution (such as y = 0) can be lost when dividing; Lesson 15.03 names this honestly. Growth and decay models are idealizations; the lesson asks students where the model stops being true.

Module 16: Area, Average Value, and Volume (Weeks 33 to 34)

Standards: MA.912.C.5.5 - MA.912.C.5.7. **Essential question:** What quantities can an integral add up, and how do you set one up from a picture?

Lessons: 16.01 area between curves; 16.02 average value of a function; 16.03 volumes by cross-sections; 16.04 volumes of revolution: disks and washers.

Watch for: Sketch first. Students must set up the slice before integrating: top minus bottom (or right minus left), radius or cross-section area as a function of the variable. The shell method is not required and is not taught.

FINALS WEEK (WEEK 35)

Review lesson (35.01): synthesizes Modules 9 to 16 and includes the cross-unit threads: velocity and acceleration (Lessons 8.02, 12.03, 15.01), reading graphs of f, f-prime, and f-double-prime into the Fundamental Theorem, and the Mean Value Theorem returning as the Mean Value Theorem for Integrals. **Review Activity** (20 points): ten mixed problems plus a one-page hand-drawn "derivative-to-integral field guide" chart, uploaded as a PDF. **Semester 2 Final Exam** (60 points): cumulative over Modules 9 to 16 with light spiral from Semester 1; hand-written in one sitting (about two hours), one PDF upload, honor line first. Part A fluency (7 items), Part B reasoning (4 items), Part C synthesis (2 items). Scored with the Semester Final Rubric: four criteria (understanding, quality of reasoning, clarity and precision, intellectual honesty) at 15 points

each. The time limit is communicated, not enforced; the honor line is the trust model. Release the worked solutions after grading closes so students can review.

ANSWER KEYS AND GRADING

The unpublished Teacher Resources module holds a key for every problem set and every Module Challenge in Modules 5 to 16, for the Semester 2 Review Activity and Final, and (in the Semester 1 kit) for Modules 1 to 8. Problem sets are scored on completion; use the keys for spot checks and for the discussion and Module Challenge. Grade Module Challenges with the shared four-criterion rubric in the Rubrics document, and keep it identical across modules so five-minute-per-student grading stays feasible. Quiz answers are in the quizzes themselves.

SEQUENCING AND SCOPE NOTES

The course follows the Florida course description in order: limits and continuity, derivatives and their applications, graph analysis, then integrals. Topics outside the standards are deliberately absent: Newton's method, the shell method, arc length, work, and hyperbolic functions. The lessons assume the Semester 1 toolkit (limits, continuity theorems, all differentiation rules, L'Hôpital's Rule, the Mean Value Theorem); several lessons cite earlier lessons by number, so students who start Semester 2 without Semester 1 will need a short bridge.

TECHNICAL NOTES

Lesson pages and module maps are hosted on GitHub Pages and embedded in Canvas by iframe; the interactive widgets and MathJax run inside the frame. Canvas-native items (quizzes, problem sets, discussions, surveys) carry their own math through Canvas's math delimiters. If a lesson frame does not load, each page offers an "open in a new tab" link.